

System Test Plan

Team3-Basyx-Editor

Version Control

Version	Date	Author	Comment
1.0	05.05.2026	Daniel Ziegler	Version 1.0
1.1	06.05.2026	Daniel Ziegler	fix table of contents
1.2	06.05.2026	Daniel Ziegler	Added test results
1.3	15.05.2026	Daniel Ziegler	Added external links, small format changes

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1. Introduction

This System Test Report (STR) documents the results of the system testing activities conducted for the BaSyx Editor Extension. The testing process was performed in accordance with the defined System Test Plan ([STP](#)) and aimed to verify that the implemented functionality meets the requirements specified in the Software Architecture Specification ([SAS](#)).

2. Scope

The scope of this report includes the evaluation of the following subsystems:

- [MOD01](#) – File Import and Validation Subsystem
- [MOD02](#) – XML Viewer and Navigation Subsystem
- [MOD03](#) – AAS Generation and Submodel Mapping Subsystem

As defined in the STP, this report does not include results related to unit testing of minor code changes or internal BaSyx backend functionality.

3 Results of Test Cases

3.1. TC01 Proper XML-Validation MOD01

Description:	The test case verifies that the function correctly validates XML, KBL, and VEC files by checking their headers and returns an appropriate error if the header is missing, invalid, or malformed.		
Unit test			
Found in:	./tests/utlis/XMLValidator.test.ts		
Tester:	Daniel Ziegler		
Date:	06.05.2026		
Test result:	PASSED		

3.2 TC02 Handle Error Messages MOD01

3.2. TC02 Handle Error Messages MOD01

ID:	<TC02>		
Name:	Handle Error Messages		
Req.-ID:	FR.01, FR.03		
Description:	The test case verifies that incorrect XML, KBL, and VEC files return consistent, accurate, and meaningful error messages.		
Test steps			
Step	Action	Expected result	Actual result
1	Click the "add Submodel Element (+)" button within a Submodel	A dialog for the type of Element opens	A dialog for selecting the element type appears
2	Select "file" and click "next"	A dialog for element details opens	A dialog for element details appears.
3	Under "Value" click "file" and add a file in the file input.	The file is added to the dialog, the ShortID is filled with the filename if empty. The MIME type is displayed correctly.	The selected file is added to the dialog, the ShortID is automatically populated and the MIME type is displayed correctly.
4	Optionally change other values including the ShortID	The changes are displayed	The changes are displayed
4	Click "save"	If the file is invalid, the file input is marked in red and an error is shown. If valid, an element is created with the correct MIME type.	For invalid files they are marked in red and an error is shown. For valid files an element with the correct MIME type is created.
Dataset:	Expected Result:		Actual Result:
1	No error		No error
2	No error		No error
3	No error		No error
4	No error		No error
5	Error: XML is not well-formed.		Error: XML is not well-formed.
6	Error: XML is not well-formed.		Error: XML is not well-formed.
7	Error: XML header is missing.		Error: XML header is missing.
8	Error: XML is not well-formed.		Error: XML is not well-formed.
9	Error: KBL_container is not a valid root element for .vec files.		Error: KBL_container is not a valid root element for .vec files.
10	No error		No error
11	No error		No error
12	Error: XML is not well-formed.		Error: XML is not well-formed.
13	Error: XML is not well-formed.		Error: XML is not well-formed.
14	Error: VecContent is not a valid root element for .kbl files.		Error: VecContent is not a valid root element for .kbl files.
15	No error		No error
16	No error		No error
Tester:	Daniel Ziegler		
Date:	06.05.2026		
Test result:	PASSED		

3.3. TC03 Display XML structure MOD02

3.3. TC03 Display XML structure MOD02

Name:	Display XML structure		
Req.-ID:	FR.04		
Description:	The test case verifies that the XML structure is correctly represented as a table of contents and that selecting entries navigates to the corresponding XML sections accurately and consistently within the viewer.		
Test steps			
Step	Action	Expected result	Actual result
1	Choose a Submodel with a XML-type file element and open it.	A list of elements in the Submodel is shown.	A list of elements in the Submodel is displayed.
2	Click on the XML-type file.	A detailed description of the Element opens on the right.	A detailed description of the Element appears on the right.
3	Click on "Visualization".	The XML preview with a table of contents and the XML content opens.	The XML preview with a table of contents and the XML content is displayed.
4	Scroll the XML content.	Only the XML content is scrolled.	Only the XML content scrolls.
5	Scroll the table of contents.	Only the table of contents is scrolled.	Only the table of contents scrolls.
6	In table of contents click on an entry.	The XML content jumps to this entry and highlights the line.	The XML content jumps to the selected entry and highlights the line.
7	In table of contents click on the triangle next to an entry.	The inner XML structure is revealed and the XML content does not jump to the entry.	The inner XML structure is revealed and the XML Content does not jump to the entry.
Dataset:	Expected Result:	Actual Result:	
1	The Visualisation works	The Visualisation works	
2	The Visualisation works	The Visualisation works	
3	The Visualisation works	The Visualisation works	
4	The Visualisation works	The Visualisation works	
Tester:	Daniel Ziegler		
Date:	06.05.2026		
Test result:	PASSED		

3.4. TC04 AAS-Creation-Wizard functions properly MOD03

3.4. TC04 AAS-Creation-Wizard functions properly MOD03

Name:	AAS-Creation-Wizard functions properly		
Req.-ID:	FR.05		
Description:	The test case verifies that the creation wizard's graphical interface correctly guides the user through the process, displaying appropriate inputs, navigation, and feedback when creating Asset Administration Shells from KBL or VEC files.		
Test steps			
Step	Action	Expected result	Actual result
1	On the slide for the Asset Administration Shells, click on the three dots and on "Create AAS form KBL/VEC"	The wizard for the AAS creation opens	The wizard for the AAS creation appears
2	Click on the FileInput and choose a KBL or VEC file.	Only eligible files can be chosen. The file is displayed.	The file is displayed.
3	Optionally choose a picture for the AAS in the second FileInput.	Only eligible files can be chosen. The file is displayed.	The file is displayed.
4	Provide or generate AAS ID and Asset ID and Type. Optionally change general Information for the AAS that will be created	The fields for the Information are easy to use and display the entered values. The "generate" button fills the corresponding textinput with a generated value.	When pressing "generate" the corresponding textinput fills.
5	Choose Information which should be used to fill the Submodels of the AAS.	Individual Information can be selected or deselected.	Individual Information can be selected or deselected.
6	Click "Create"	The AAS is created with the provided Information.	The AAS is created using the provided Information.
Dataset:	Expected Result:	Actual Result:	
1	The corresponding AAS is created	The corresponding AAS is created	
3	The corresponding AAS is created	The corresponding AAS is created	
Tester:	Daniel Ziegler		
Date:	06.05.2026		
Test result:	PASSED		

3.5. TC05 Extraction of Information for AAS-Generation MOD03

Test case			
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3.5. TC05 Extraction of Information for AAS-Generation MOD03

Req.-ID:	FR.06
Description:	The test case verifies that AAS information is correctly extracted from a given file, ensuring all relevant elements are accurately identified, parsed, and made available for further processing.
Unit Tests	
Found in:	./tests/utlis/KblVecUtils/KblVecDataPointTreeUtils.test.ts ./tests/utlis/KblVecUtils/KblVecRequiredFieldUtils.test.ts ./tests/utlis/KblVecUtils/KblVecSubmodelGenerationUtils.test.ts
Tester:	Daniel Ziegler
Date:	03.05.2026
Test result:	ERROR WITH: prefers File-Name over XML ERROR WITH: interprets <aas>-tags correctly ERROR WITH: reads description data point to only description
Tester:	Daniel Ziegler
Date:	06.05.2026
Test result:	PASSED